Fixed Length Walk Queries

Time limit: 1.00 s **Memory limit:** 512 MB

You are given an undirected graph with n nodes and m edges. The graph is simple and connected.

You start at a specific node, and on each turn you must move through an edge to another node.

Your task is to answer q queries of the form: "is it possible to start at node a and end up on node b after exactly x turns?"

Input

The first line contains three integers n, m and q: the number of nodes, edges and queries. The nodes are numbered $1, 2, \ldots, n$.

After this, there are m lines which describe the edges. Each line contains two integers a and b: there is an edge between nodes a and b.

Finally, there are q lines, each describing a query. Each line contains three integers a, b and x.

Output

For each query, print the answer (`YES` or `NO`) on its own line.

Constraints

- $2 \le n \le 2500$
- $1 \le m \le 5000$
- $1 \le q \le 10^5$
- $0 \le x \le 10^9$

Example

Input

```
4 5 6
1 2
2 3
1 3
2 4
3 4
1 2 2
1 4 1
1 4 5
2 2 1
2 2 2
3 4 8
```

Output

```
YES
NO
YES
NO
YES
YES
```

Explanation:

- In query 1, a possible route is $1 \to 3 \to 2$.
- In query 3, a possible route is $1 \to 3 \to 2 \to 1 \to 3 \to 4$.
- In query 6, a possible route is $3 \to 4 \to 2 \to 3 \to 4 \to 2 \to 1 \to 3 \to 4$.